Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2000-II-Q01] | Part A (Q1-Q5)

Answer: 007

1. Reciprocal logarithms convert the expression to $2\log_{2000^6} 4 + 3\log_{2000^6} 5 = \log_{2000^6}(4^2 \cdot 5^3)$.

2. Since $4^2 \cdot 5^3 = 2000$, the value is $1/6$ and the answer is 7.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2003-I-Q04] | Part A (Q1-Q5)

Answer: 012

1. The logarithm domains make both sine and cosine positive, and the first relation gives their product $1/10$.

2. Their squared sum is $1 + 2/10 = 12/10$. Taking the logarithm gives exactly $(\log_{10} 12 - 1)/2$, so $n = 12$.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2005-II-Q05] | Part A (Q1-Q5)

Answer: 054

1. Put $t = \log_a b$, so $t + 6/t = 5$. Factoring $t^2 - 5t + 6$ yields $t = 2, 3$, hence $b = a^2$ or a^3.

2. Squares allow $a = 2, \ldots, 44$, giving 43 pairs; cubes allow $2, \ldots, 12$, giving 11. The families do not overlap for $a \geq 2$, so total 54.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2006-II-Q02] | Part A (Q1-Q5)

Answer: 893

1. Triangle inequalities require $\log(75/12) < \log n < \log(12 \cdot 75)$, equivalent to $25/4 < n < 900$. These also ensure positivity and the remaining inequality.

2. Thus $n = 7, \ldots, 899$, giving 893 values.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2009-II-Q02] | Part A (Q1-Q5)

Answer: 469

1. Raise each given equality to its own logarithmic exponent. The terms become $27^{\log_3 7} = 7^3$, $49^{\log_7 11} = 11^2$, and $(\sqrt{11})^{\log_{11} 25} = 5$.

2. The sum is $343 + 121 + 5 = 469$.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2010-I-Q03] | Part A (Q1-Q5)

Answer: 529

1. Take the positive xth root after substitution: $x^{3/4} = 3x/4$. Thus $x^{-1/4} = 3/4$ and $x = 256/81$.

2. Then $y = 64/27$, so $x + y = 448/81$ in lowest terms. The answer is 529.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2010-II-Q05] | Part A (Q1-Q5)

Answer: 075

1. Put $a = \log_{10} x, b = \log_{10} y, c = \log_{10} z$. Then $a + b + c = 81$ and $ab + ac + bc = 468$.

2. Thus $a^2 + b^2 + c^2 = 81^2 - 2 \cdot 468 = 5625$, whose positive square root is 75.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2013-II-Q02] | Part A (Q1-Q5)

Answer: 881

1. Undo the logarithms outward to inward: the middle logarithm is one, the innermost logarithm is 2^a, and therefore $b2^a = 1000$.

2. Since $1000 = 2^3 \cdot 125$, positive a is $1, 2, 3$, giving $(a, b) = (1, 500), (2, 250), (3, 125)$.

3. The sums are distinct and total $501 + 252 + 128 = 881$.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2016-II-Q03] | Part A (Q1-Q5)

Answer: 265

1. Put $u = \log_5 x, v = \log_5 y, w = \log_5 z$, with sum s. Exponentiating the outer logs gives $5^s + u = 35$, $5^s + v = 84$, and $5^s + w = 259$.

2. Add: $3 \cdot 5^s + s = 378$. The left side is strictly increasing, and $s = 3$ solves it, so it is the only solution.

3. Then $(u, v, w) = (-90, -41, 134)$. These define positive x, y, z and valid original logarithm arguments. The requested absolute sum is $90 + 41 + 134 = 265$.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2018-I-Q05] | Part A (Q1-Q5)

Answer: 189

1. The first equation with its positive argument is equivalent to $(2x+y)^2 = x^2 + xy + 7y^2$, factoring as $(x-y)(x+2y) = 0$. Neither branch allows $y = 0$.

2. The second equation similarly gives $K = 6(x/y)^2 + 2(x/y) + 1$. The branches $x/y = 1, -2$ yield $K = 9, 21$.

3. Both occur with all log domains valid: choose $y > 0$ in the first branch, $y < 0$ in the second. Their product is 189.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2020-I-Q02] | Part A (Q1-Q5)

Answer: 017

1. Use $t = \log_2 x$ to rewrite the terms as $(1+t)/3, t/2, t$. Their relation is $t^2/4 = t(1+t)/3$.

2. $t = 0$ produces $1/3, 0, 0$, which has no positive common ratio. For $t \neq 0$, $3t = 4 + 4t$, giving $t = -4$.

3. Therefore $x = 1/16$; the terms are $-1, -2, -4$, so the ratio is indeed 2. The result is 17.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2020-II-Q03] | Part A (Q1-Q5)

Answer: 103

1. Logarithm bases impose $x \neq 0, -3$. Changing to one fixed base and cancelling $\log 3 / \log 2$ gives $20/x = 2020/(x+3)$.

2. Thus $20x + 60 = 2020x$, so $x = 3/100$, which satisfies both base restrictions. The result is 103.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2022-II-Q04] | Part A (Q1-Q5)

Answer: 112

1. Call the common value t. Then $\ln(22x) = t\ln(20x)$ and $\ln(202x) = t\ln(2x)$.

2. Subtract the first equality from the second to eliminate $\ln x$: $\ln(101/11) = t\ln(1/10)$. Thus $t = \log_{10}(11/101)$.

3. The given existence is consistent: $(1-t)\ln x = t\ln 20 - \ln 22$ determines a positive x, and neither excluded value satisfies this equation. The coprime pair is $(11, 101)$, giving 112.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2023-I-Q02] | Part A (Q1-Q5)

Answer: 881

1. Put $t = \log_b n \geq 0$. The first relation becomes $\sqrt{t} = t/2$, whose only possibilities are $t = 0, 4$.

2. The second relation is $bt = 1 + t$, ruling out $t = 0$. Hence $t = 4$ and $b = 5/4$.

3. Therefore $n = (5/4)^4 = 625/256$. The powers of 5 and 2 are coprime, so the sum is 881.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2024-I-Q02] | Part A (Q1-Q5)

Answer: 025

1. Extracting the powers gives $x \log_x y = 10$ and $4y \log_y x = 10$.

2. Multiply: change of base yields $(\log_x y)(\log_y x) = 1$, so $4xy = 100$ and $xy = 25$.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2024-II-Q04] | Part A (Q1-Q5)

Answer: 033

1. Put $a = \log_2 x, b = \log_2 y, c = \log_2 z$. The equations become $a - b - c = 1/2, b - a - c = 1/3, c - a - b = 1/4$. Their sum gives $a + b + c = -13/12$.

2. Therefore $(a, b, c) = (-7/24, -3/8, -5/12)$, and $|4a + 3b + 2c| = 25/8$. The required sum is 33.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2025-II-Q04] | Part A (Q1-Q5)

Answer: 106

1. Change base and factor the exponents. The kth factor becomes $\frac{k-1}{k-2}\frac{k+1}{k+2}\frac{\log(k+1)}{\log k}$.

2. The three products telescope to $62/2 = 31$, $5/65 = 1/13$, and $\log 64/\log 4 = 3$. Their product is $93/13$, so the requested sum is 106.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2000-I-Q09] | Part B (Q6-Q10)

Answer: 025

1. Put $a = \log_{10} x, b = \log_{10} y, c = \log_{10} z$, and $L = \log_{10} 2$. The equations become $(1 - a)(1 - b) = L$, $(1 - b)(1 - c) = L$, $(1 - c)(1 - a) = 1$.

2. Since $L \neq 0$, the first two yield $a = c$. Therefore $1 - a = 1 - c = \pm 1$. Correspondingly $1 - b = \pm L$, with matching signs.

3. Thus $y = 10^{1-L} = 5$ or $10^{1+L} = 20$, and their sum is 25.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2002-I-Q06] | Part B (Q6-Q10)

Answer: 012

1. Let $p = \log_{225} x, q = \log_{64} y$, with $p, q \neq 0$. Then $p + q = 4$ and $1/p - 1/q = 1$, yielding $p^2 - 6p + 4 = 0$.

2. The two p-values sum to 6, and the q-values to 2. Hence the product is $225^6 \cdot 64^2 = 30^{12}$, so the required logarithm is 12.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2005-I-Q08] | Part B (Q6-Q10)

Answer: 113

1. Put $y = 2^{111x} > 0$, converting the equation to $y^3 - 8y^2 + 16y - 4 = 0$. The three stated real x roots correspond bijectively to three positive y roots.

2. By Vieta their product is four, so the x-sum is $\log_2(4)/111 = 2/111$. The answer is 113.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2011-I-Q09] | Part B (Q6-Q10)

Answer: 192

1. Positivity permits exponentiation and squaring: $(24\sin x)^3 = (24\cos x)^2$, hence $24\sin^3 x = \cos^2 x$.

2. With $s = \sin x$, this is $24s^3 + s^2 - 1 = (3s-1)(8s^2+3s+1) = 0$. The quadratic has negative discriminant, so $s = 1/3$. It gives valid base $8 \neq 1$ and positive argument.

3. Then $\cot^2 x = (1 - 1/9)/(1/9) = 8$, and the result is 192.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2012-I-Q09] | Part B (Q6-Q10)

Answer: 049

1. Write $x = 2^a, y = 2^b, z = 2^c$, and denote the nonzero common value by t. Then $b + 1 = ta/2, c + 2 = t(a + 1)/2, b + c + 3 = t(4a + 1)$.

2. Add the first two expressions and compare with the third: $t(2a + 1)/2 = t(4a + 1)$. Since $t \neq 0$, one gets $a = -1/6$.

3. The first two relations now imply $c + 2 = -5(b + 1)$, or $5b + c = -7$. Hence $xy^5z = 2^{-43/6}$ and $p + q = 49$.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2013-I-Q08] | Part B (Q6-Q10)

Answer: 371

1. The domain is $[1/(mn), m/n]$, so its length condition is $2013(m^2-1) = mn$.

2. As m is coprime to m^2-1, one must have $m \mid 2013 = 3 \cdot 11 \cdot 61$. Conversely every such divisor greater than one yields integer $n = 2013(m - 1/m)$.

3. The sum is $2014m - 2013/m$, strictly increasing for positive m. The least permitted divisor is three, giving $n = 5368, m+n = 5371$ and remainder 371.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2017-II-Q07] | Part B (Q6-Q10)

Answer: 501

1. The domain requires $x > -2$ and $kx > 0$. On this domain the equation is $kx = (x+2)^2$, or $x^2 + (4-k)x + 4 = 0$.

2. For each $k < 0$, the allowed interval is $(-2, 0)$; the line kx strictly decreases while $(x+2)^2$ strictly increases, with opposite endpoint order, giving exactly one crossing. All500 negative values work; $k = 0$ is undefined.

3. For $k > 0$, one needs positive roots. Discriminant $k(k-8)$ is negative below8, zero at8 with double root2, and positive above8 with two positive roots. Only $k = 8$ contributes, so the total is501.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2019-II-Q06] | Part B (Q6-Q10)

Answer: 216

1. Put $y = \log_b x$. Change of base in the second equation gives $\log_b y = y/54$. The first becomes $y/2 + \log_b y = 56/3$.

2. Thus $y/2 + y/54 = 56/3$, so $y = 36$. Then $\log_b 36 = 2/3$, giving $b = 36^{3/2} = 216$. This value fixes a unique y, hence a unique x.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2026-I-Q06] | Part B (Q6-Q10)

Answer: 441

1. Put $t = \log_{2026} x$ and take logarithms. This yields $t^2 - 20t - 20\log_{2026} 26 = 0$, with two distinct real roots and therefore two positive x values. Vieta gives root sum 20.

2. The product is $2026^{20} = 2^{20}1013^{20}$. Trial division by primes up to 31 verifies 1013 prime, so the divisor count is $(20+1)^2 = 441$.

Source / 来源

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2017-I-Q14] | Part C (Q11-Q15)

Answer: 896

1. Put $c = \log_2 a > 0$. Exponentiating the outer first logarithm and multiplying by c gives $\log_2(24/c) = 128c + c2^{128c}$. The left side is strictly decreasing and the right strictly increasing, so there is at most one positive solution.

2. The value $c = 3/64$ works: the left is $\log_2 512 = 9$, and the right is $6 + 3 = 9$. The second equation then gives $x = a^{a^{256}} = (2^{3/64})^{4096} = 2^{192}$, a positive integer.

3. Modulo8 the result is0. Modulo125, Euler gives $2^{192} \equiv 2^{-8} \equiv 6^{-1} \equiv 21$. Solving these two congruences gives $x \equiv 896 \pmod{1000}$.

Source / 来源